Monte Carlo Skill Estimation for Darts

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***Abstract*—In physical games, like darts, the ability of a player to accurately execute an intended action has a significant impact on their success. Determining this execution precision, or skill, for players is thus an important task. Knowledge of skill can be used for player feedback, computer-aided strategy decisions, game handicapping, and opponent modeling. Challenges to estimating player ability include getting precise feedback on executed actions as well as performing the estimation in a natural and user-friendly way. A previous method for estimating skill in darts overcomes the first challenge, but falls short on the second, requiring players to throw 50 darts at the center of the dartboard, which is not a common target in most darts games. In this paper we present an extension of this previous method that enables skill to be estimated when darts are aimed anywhere, not just the center of the dartboard. This method is then utilized to develop a much more efficient and adaptive skill estimation method which requires far fewer darts than the previous method. Experimental results demonstrate the advantages of the proposed approach and additional possible applications are discussed.**

***Index Terms*—Player modeling, skill estimation, Monte Carlo methods**

I. Introduction

For many physical game domains an agent's skill level, the accuracy with which they can execute intended actions, is a key factor of their performance and ability. Understanding of player skill is crucial to a variety of applications related to games. These include handicapping, player modeling, computer assisted strategy, as well as many others. This paper focuses on the task of estimating a player's skill in one such game, the game of darts. In many games, pseudo-metrics are used to compare the skill levels of different players. In chess, players are ranked with the Elo system [1]. In golf, players are given a handicap based on previous scores. Both of these metrics are useful for comparing players to each-other, but fail to describe the underlying execution ability of the players. It would be very difficult to know in which areas a golfer might improve by only knowing his handicap. While Elo, score-handicapping, or other established rating methods could be applied in darts, we seek to provide a much more descriptive and accurate representation of execution ability by understanding the distribution of dart throws rather than just a players average score or weighted win-loss ratio. Doing this provides a stronger foundational framework for the applications mentioned above.

As with many other games, darts has specific factors that prohibit simple and straightforward estimation of an agent's skill. The key issue in the game of darts is that, it generally infeasible to measure the exact result of a player's throw.[1] If this precise information were known, an estimate of the agent's skill could be attained in a straightforward manner, seen in (1). The only information that is available after a typical darts throw, however, is the score for that throw. Each score represents a variety of different locations where a dart could have landed, and thus, information about the throw is lost.

Skill estimation methods must overcome this information loss to be successful. Additionally, to be practical it is best if they can be used in natural, user-friendly ways. This means that we would prefer to be able to estimate skill from actions of players playing real games and that it would be better if the estimators require fewer observations to make an accurate skill estimate.

A previous method [2] was able to successfully overcome the information loss in order to estimate skill in darts. They estimated an agent's skill level through an adaptation of the expectation maximization (EM) algorithm, using only the score information of the dart throws rather than their location. However, their approach is limited as it relies on certain radial symmetries of the dartboard and can only work for throws aimed at the center of the dartboard. In most popular darts games, this is not a common target. Thus, to estimate a player's skill, they must go through a separate estimation process, throwing 50 darts at the center of the board.

In this paper we present an extension of their approach that allows for skill to be estimated regardless of the target for the throws. This added flexibility allows for the skill of players to be estimated during a normal game, as long as the aiming point of each throw is known. In addition, a variety of other applications to improve player experience in darts can be explored. One such application is explored in this paper, a more efficient skill estimation procedure. This procedure can be used when the goal is simply to estimate a players skill in a similar manner to the 50 darts-at-the-center procedure of the previous work. Our new adaptive procedure is shown to work much more efficiently.

[1]There are currently a few applications that allow players to take an image of the dartboard and somewhat accurately be provided with the corresponding scores, however obtaining the actual precise location of the dart can currently only be done with very expensive equipment.

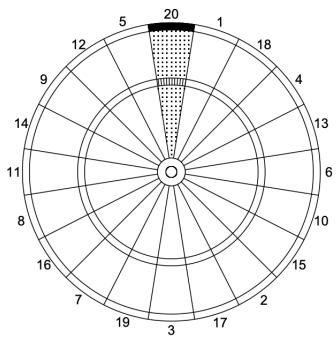

Fig. 1. The standard dartboard. The different regions are represented on the 20 wedge, dotted being single, solid being double, and striped being triple

This paper proceeds as follows: first, relevant background is provided in Section II. Second, our approach for estimating skill with arbitrary aiming points is presented in Section III. Next, given that any aiming point could be used to estimate skill, the question of "which aiming point is best for estimating skill?" is investigated in Section IV. The results of that investigation are used in Section V to develop a novel adaptive skill estimator that is shown experimentally to outperform the previous method. We conclude with a summary and discussion of potential future applications of these new ideas.

While the research in this paper deals within the domain of darts, the findings with respect to the modeling of information loss due to measuring procedures as well as the use of Monte Carlo to empirically improve estimates of execution skill can be applied to other physical game domains with similar properties.

II. Background

This section provides necessary background information to understand the remainder of the paper.

A. Darts

The family of darts games, of which there are a large variety, all consist of throwing small pointed projectiles (the darts) at a circular target board (the dartboard). Typically, the darts stick in the dartboard and each dart receives a score depending on the region of the dartboard where it stuck.

The standard dartboard consists of 20 distinct wedge-shaped scoring regions as well as two concentric circular bands of 'multiplying' regions, a triple region and a double region. The score for a dart landing within one of the multipliers is the product of the scoring region and the multiplier. Two additional scoring region, the bullseye (25 points) and double-bullseye (50 points), are in the very center of the board. The standard dartboard is shown in Fig 1, with the score for each wedge-shaped region shown on the outside of that region.

While the various different dart game rules are distinguished by distinct win-conditions and objectives, in the most common games players take turns throwing darts at the board (usually in batches of 3) and recording the scores of each throw. The most popular of these games among professional and amateur players is 501 [3]. In 501 darts, players start with 501 points and each dart thrown then subtracts that score from the total. The goal of the game is to reach exactly zero, with the constraint that your first and last scoring darts must be within a double region[2]. This version of the game incorporates aspects of maximizing score, as well as pursuing specific score combinations.

The main physical skill necessary for a player to be successful at darts is the ability to consistently throw the dart close to their target. We refer to this as execution skill, execution precision, execution accuracy, or often simply skill, and the main focus of this paper is on its estimation. The main mental skill a player needs is the ability to determine the best place to aim, given their execution precision and the scores they desire for the game being played. It is important to note that the purpose of our skill estimators is to measure a players execution ability and not their planning/strategic ability.

B. Modeling Darts: Notation and Terminology

In this section the mathematical notation and terminology used to reason about darts are presented.

The aiming point, or target, of a player's throw will be indicated by a point $\mu = (\mu_x, \mu_y)$. Given that a player is aiming at target point μ, the actual result $X = (x, y)$ of each dart throw is modeled as a draw from a 2-dimensional symmetric Gaussian distribution, as follows:

$$X \sim \mathcal{N}(\mu, \sigma^2 I)$$

where I is a 2×2 identity matrix.

The standard deviation σ of this distribution corresponds to the player's execution precision and for darts is measured in millimeters. While the player may change aiming points (μ) throughout the game, we assume that a player's skill level won't change during the course of a game and thus treat σ as a fixed value.

1) Skill levels: The following scale with approximate descriptors is provided to help give some better intuition about the numerical scale of σ:

- $\sigma = 10$: (Elite Player) Will hit the bullseye 90% of the time when aiming at it
- $\sigma = 63$: (Amateur Player) Will hit within the area defined by the edge of the triple region 90% of the time when aiming at the center
- $\sigma = 107$: (Beginner Player) Will hit the board 90% of the time when aiming at center

In this paper we consider players up to $\sigma = 170$ for which the entire board represents one standard deviation, meaning a player at this skill level will hit the board 68% of the time.

[2]Most professional games do not require the first score to be a double, to speed up play

2) Scoring: As mentioned previously, it is typically impractical to observe the exact $X = (x, y)$ landing location for a dart throw. However, what is easily observed is the score for a dart, which is determined by the scoring region it landed in. The score for a dart will be represented as another random variable $Z = score(X)$, where $score$ is a function that generates the score for a given location X on the dart board. We will indicate the set of all unique possible scores as S, with elements s_i[3].

C. Skill estimation

We are now prepared to present the problem which is the main focus of this paper: estimating a player's execution skill. Our task is to estimate σ, given a set of n observed scores $[Z_0, \dots Z_n]$, with their corresponding aiming points $[\mu_0, \dots, \mu_n]$. A solution to this problem will take in the scores and aiming points and produce a σ value. To be useful, the method used needs to be accurate, converge quickly as n increases, and give consistent estimates for the unknown variable σ. In this paper the accuracy of an approach is measured by its ability to, given a certain number of scores, provide an estimate close to the maximum likelihood estimator of Section II-C1. The convergence of an approach is measured by how its accuracy increases as the number of observations increases. Consistency is measured by the variance of the estimate across different sets of n samples from the same distribution X. As an example, if approach A gives a 7% error after throwing 30 darts, where approach B gives a 9% error, approach A would be more accurate. However, approach B would be more consistent if it was found that the inner quartile ranges were $(2\%, 18\%)$ and $(5\%, 11\%)$ respectively, across many repetitions of observing 30 darts. An ideal approach would be more accurate and also more consistent for each possible size dataset.

1) Baseline method: If the exact X dart locations could be observed, a straightforward maximum likelihood estimator (MLE) would be applicable. Equation (1) provides the MLE calculation for σ, based on a set of throws $X = [x_0, \dots x_n]$ and aiming points of size n.

$$\sigma^2 = \frac{1}{2n} \sum |\mu_i - x_i|_2^2 \tag{1}$$

This approach is accurate, consistent, and has a high rate of convergence. However, it cannot be used to solve our problem in general because the precise X values are not known. In our experimental framework the X values are known and so the MLE estimate will be used as a best-case baseline and the different approaches will be compared to this to evaluate their performance.

2) Existing Approach: A method for solving the skill estimation problem in darts was presented in [2]. This method is based on an adaptation of the EM algorithm first introduced by [4] and consists of an expectation (E) step and a maximization (M) step. The E step relies on the radial symmetry of the dartboard to provide expected distances from the center of the board, given scores, and as such it can only be used when μ is the center of the dartboard. The M step uses an MLE estimator for σ given the expected distances computed during the E step. The main success of this method is that it is able to be run with only the dart scores as input. It was shown in [2] to be accurate and converge relatively quickly. However, it still has the major drawback of only making estimates of a player who is aiming at the center of the dartboard.

When the target μ is a location other than the center, the expectation function proposed by [2] becomes too difficult to evaluate analytically. In this paper we propose a method to overcome this difficulty by replacing their E step with a Monte Carlo-based estimate of the expected distances. Augmenting EM with Monte Carlo estimation in this general way is called *Monte Carlo Expectation Maximization* (MCEM), and was first introduced in [5]. MCEM provides a numeric solution to problems where the traditional EM algorithm is infeasible due to lack of a well defined estimation function, by introducing a Monte Carlo calculation in its place. The specifics of our approach are detailed in the next section.

III. Monte Carlo EM for Darts

Our MCEM approach for skill estimation in darts overcomes the shortcoming of the previous approach and allows throws with any aiming points μ to be input. While the main goal of this approach was to be more widely applicable, we show that it also results in performance improvements on certain tasks.

For simplicity of exposition, we present the MCEM algorithm for darts skill estimation assuming a common aiming point μ for all the throws. A simple modification that can account for distinct μ values will be discussed later. Given a set of scores $Z = [z_0, \dots, z_n]$, an aiming point μ, and an initial skill estimate σ_t, the MCEM approach proceeds iteratively as follows, alternating E and M steps to produce a new skill estimate σ_{t+1} until termination:

1) **Estimation (E) Step:**
   a) Generate the set of Monte Carlo samples $Y = [y_1, \dots y_M]$, where each $y_j \sim \mathcal{N}(\mu, \sigma_t^2 I)$.
   b) For each unique scoring region $s_k \in S$ the mean squared distance across all points in Y with that score to μ is computed. This distance is called m_k and its computation is given by equation 2, assuming there are l samples from Y that have a score of s_k.

$$m_k = \frac{\sum_{y_j \in s_k} |\mu - y_j|_2^2}{l} \tag{2}$$

   Thus, for each possible score s_k, m_k represents the expected squared distance from a dart with that score to μ, assuming an execution skill of σ_t.
   c) A set of expected squared distances $D = [d_0, \dots, d_n]$, one for each throw, is now generated, where each d_i is the m_k expected squared distance value corresponding to z_i, the score of the i-th

[3]Note that different areas on the board can result in the same score. For example, a score of 12 could be the result of hitting the 12 region, or the double 6 region, or the triple 4 region

observed dart throw. In other words, $d_i = m_k$ if $z_i = s_k$.

2) **Maximization (M) Step:**

a) A maximum likelihood estimator produces σ_{t+1} from D. This estimator is shown in Equation 3.

$$\sigma_{t+1}^2 = \frac{\sum_{d_i \in D} d_i}{2n} \tag{3}$$

b) The procedure returns to the E step, using σ_{t+1} as the skill estimate, until desired convergence.

The MCEM algorithm can be easily modified to account for multiple aiming points μ. The change is simple, the set of samples Y will need to be generated for each unique μ, giving distinct m_k values depending on the aiming point. Each d_i will be calculated by first checking the aiming point, then using the corresponding m_k expected squared distance value.

For comparison, in [2] the D expected squared distance values are computed analytically by integrating the noise distribution over the regions of dartboard. The D values using their approach are the true values, assuming the underlying noise distribution is correct. In the MCEM approach these values are estimated through Monte Carlo sampling, as their analytic computation is impractical, given that the dartboards radial symmetry cannot be exploited. For full details on the previous approach, see [2].

A. Evaluation

To evaluate the effectiveness of the MCEM approach, a number of experiments were run. Fig 2 compares the percent difference between D values computed with MCEM and the true values computed by the previous approach [2], as a function of M, the number of Monte Carlo samples. This shows that the MCEM approach only has a very small level of error when estimating D as compared to the true analytical solution. Since the M step is the same in both approaches, it is natural to expect that this means their performance will be virtually identical, assuming enough samples are used.

Figure 3 compares the accuracy of skill estimation with each approach, where MCEM uses $M = 15,000$ Monte Carlo samples. This experiment only included throws aimed at the center, so that both methods could be applied. The accuracy is percentage error from the MLE estimate using the actual dart locations, averaged across 80 different experiments. It is clear that the new MCEM approach converges almost identically to the previous approach, meaning no performance is being sacrificed.

The running time of our method scales linearly with 1) M, the number of Monte Carlo samples, 2) T, the number of EM iterations, and 3) U, the number of distinct targets in the data, yielding a complexity of $\mathcal{O}(MTU)$. With $M = 15,000$, $T = 10$, and $U = 1$, a skill estimate can be produced on a standard laptop in 0.26 seconds. This is sufficiently fast to be very useful. In most darts games there is a limited number of potential targets in the data, and so skill estimation on data from these games could also be obtained very quickly. As will be shown, the benefits of being able to estimate execution skill for any aiming point make any increased costs well worth it.

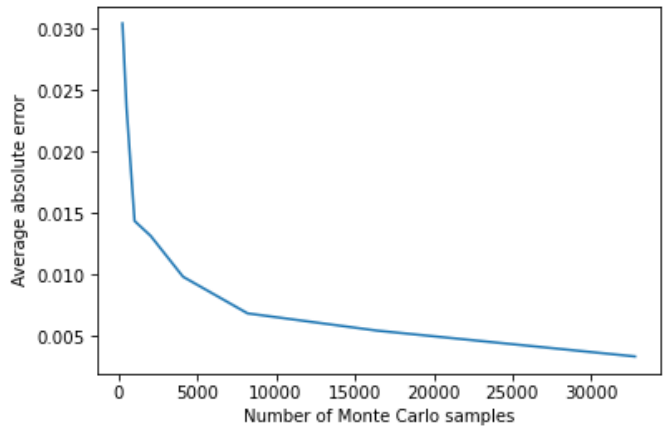

Fig. 2. Average percentage difference in D values for each Monte Carlo size

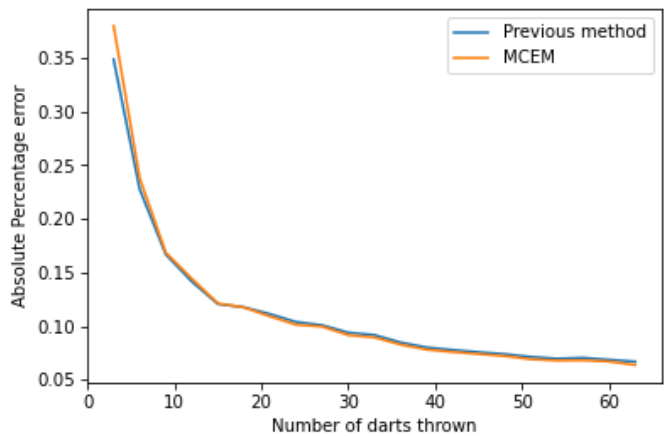

Fig. 3. Average error in predictions given a certain number of darts thrown

IV. Optimal Target for Estimation

With the MCEM approach in place, a number of different applications become possible. In the remainder of this paper we will explore one of these. As discussed previously, the new MCEM approach can be used to estimate skill over the course of a normal darts game. However, this might not be the most efficient way to provide a player with a skill estimate, if the goal is simply obtaining an accurate estimate of a player's skill in as few darts as possible. For the remainder of this paper we turn our focus to the design of a procedure that will tell a player where to aim and then estimate skill based on the resulting throws. For comparison, the procedure proposed in [2] requires a player to throw 50 darts at the bullseye in order to estimate their skill. The flexibility of our MCEM approach enables us to consider other aiming points besides the bullseye. With this in mind, we begin with a simple question: "Where should a skill estimation procedure direct a player to aim in order to most quickly obtain an accurate skill estimate?"

Due to the asymmetry of the scoring locations on the dartboard, there is potential for targets where the resulting scores give different amounts of information about a player's execution skill. For example, it may be more difficult to estimate a player's skill level if they aim at the double 20 region on the edge of the board, as many of their darts will end up off the board, giving little information about their distribution.

Naturally, we would expect our MCEM estimator to converge differently for different target locations. Intuitively, the best target would minimize the information lost between converting a dart location to its score. This concept is equivalent to minimizing the entropy of a dart's location across a scoring region.

To compute the optimal target for estimation we propose an entropy calculation for each aiming point, given an execution skill level. Different execution skill levels can yield different entropy values for the same target, since the distribution over dart locations, and thus scores, will differ.

A. Entropy Calculation

The entropy of the distribution of dart distances within a scoring region for a given target and execution skill level is computed as follows.

1) The set of possible distances from the target is discretized into evenly sized buckets. The set of all distance buckets will be referred to as B. Each possible distance from μ gets mapped to a unique bucket $b_i \in B$.
2) To estimate the distribution over distances for each score resulting from target μ and skill level σ, a large number of locations are sampled from $\mathcal{N}(\mu, \sigma^2 I)$. The score and distance from μ for each sample's location are computed, and the count of the corresponding score-distance bucket is incremented.
3) Given the set of samples, the empirical frequency $P(b_i|s, \mu, \sigma)$ of each distance bucket b_i for each score s can be computed as the fraction of samples with that score that are in that distance bucket.
4) The empirical frequencies $P(b_i|s, \mu, \sigma)$ are used to calculate the entropy of the distribution over scores for each distance, as shown in Equation 4.

$$e(s, \mu, \sigma) = -\sum_{b_i \in B} P(b_i|s, \mu, \sigma) \log P(b_i|s, \mu, \sigma) \quad (4)$$

In this equation $e(s, \mu, \sigma)$ is the estimate of the entropy of distances across the scoring region for score s, for the given μ and σ. This value can be thought of as the amount of distance information lost when converting from a set of points within the scoring region to the score s. Lower values represent less information lost, higher represents more.

5) The average amount of information lost from one throw at a given target, $E(\mu, \sigma)$, can now be calculated. This is done by computing the weighted average entropy value (e) across all scoring regions. Equation (5) shows this calculation.

$$E(\mu, \sigma) = \sum_{s \in S} e(s, \mu, \sigma) P(s|\mu, \sigma) \quad (5)$$

Here $P(s|\mu, \sigma)$ indicates the probability of hitting the scoring region s when aimed at target μ and using execution skill σ. This can also be estimated empirically from the samples generated in step 2, by counting the number of samples that have score s.

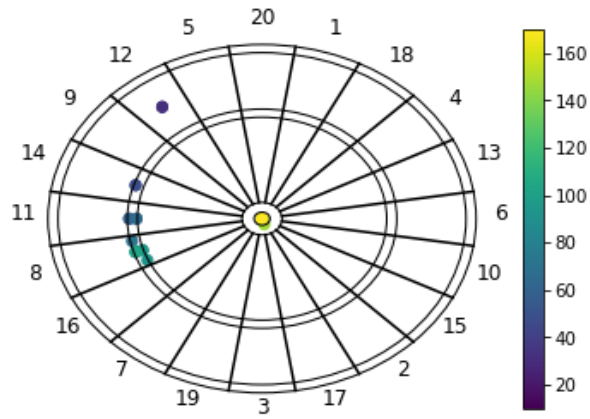

Fig. 4. Minimum entropy aiming points for 30 σ values

B. Minimum entropy targets

To determine the optimal aiming point for skill estimation for a given execution skill level, the previous entropy was computed at each of a set of points spanning the entire dartboard. This process was repeated for 30 evenly spaced σ values in the interval $[10, 170]$.

Fig 4 shows the minimum entropy points for the different execution levels. The different execution levels are shown in different colors. Note that 21 of these skills lie overlapping at the center of the board. These target locations, which we refer to as the true minimum entropy targets, will be useful for comparison.

We return to the original motivating question for this section, "What is the best aiming point overall?" This will depend upon which execution skill levels we are interested in. Assuming an even distribution across skill levels, each potential target can be evaluated by comparing how far it differs in entropy from the true minimum for each respective skill level. The overall best single target is the point that minimizes the sum of these entropy differences across all σ values.

The single best aiming point computed in this manner lies directly in the center of the triple 11, to the left of the bullseye. Figure 5 displays how this aiming point compares to the true minimum entropy aiming point for each skill level. For comparison, Figure 5 also includes a comparison of the center of the dartboard to the true minimum at each skill level.

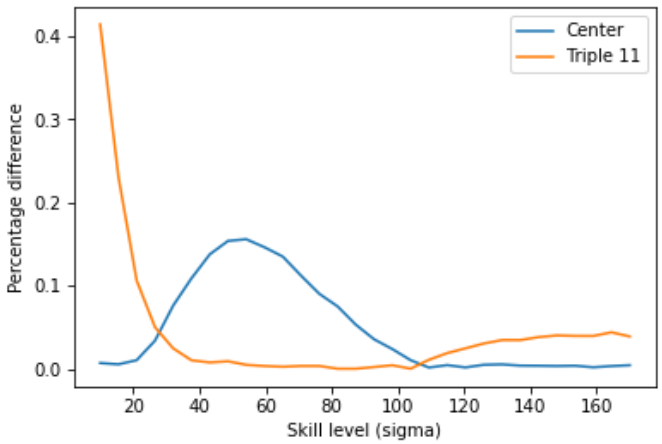

Fig. 5. Entropy difference from the respective minimum across skill levels

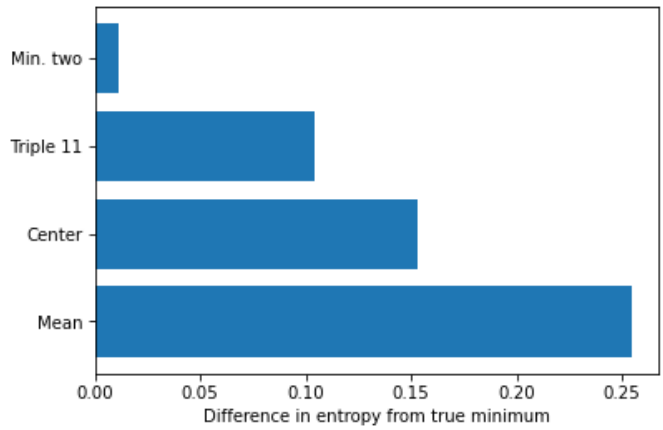

Fig. 6. Average Entropy across all skill levels (measured in relation to the true minimum)

This suggests that due to minimized information loss, making estimations while aiming at the triple 11 will result in better performance. These calculations could easily be repeated to determine the best single target for players of a given skill range or other distribution over skill levels.

V. An Adaptive Estimation Approach

An examination of Figure 5 reveals that both the center and triple 11 targets have skill levels for which they are suboptimal. This raises the possibility of an adaptive skill estimation method where the target is specified based on the current skill level estimate in order to always be using a lower entropy target and more quickly refine the skill estimate.

We first consider an adaptive approach that would utilize two targets. An exhaustive search over all pairs of targets was employed to find the pair that collectively (looking at the minimum between the two) provide the minimum entropy across all skill levels. This optimal pair ended up being the overall minimum, triple 11, along with the center of the board.

Fig 6 shows the average (across all skill levels) entropy difference from the true minimum entropy at each skill level for several different targets. First, the average for a single point selected at random, second, for the center point, third, for the the triple 11, and finally, for the minimum of triple 11 and center at each point. We see that having two targets and choosing the better of the two leads to a huge average entropy reduction from even the best single point target. We can also see that we are very close to the true minimum (using the optimal target for each execution skill) with only two targets. For this reason we chose to have the adaptive method choose between just the two targets: triple 11 and center.

Figure 8 shows the error of each constant target as a function of skill level, and the adaptive method was set up to select the target with minimum error at each skill level. This resulted in the adaptive method selecting the triple 11 when the current estimate is in the interval $[14, 74]$, and the center otherwise.

This adaptive method can be used in skill estimation settings where the the player whose skill is being estimated is told where to aim. This will allow us to compare the adaptive method with the single target methods aiming at the center [2], and the optimal single point, the triple 11. The experimental results in the next section will demonstrate that the adaptive method is superior in all evaluation dimensions: it is most accurate and converges quickly, and it also is the most consistent.

VI. Experimental Evaluation

To demonstrate the effectiveness of the proposed skill estimation methods, extensive experiments were carried out. It is important to note that this estimation method is intended to be used in the real world with real darts players, for future work we intend to apply the technique in real darts games, however to analyze the usefulness and convergence of the algorithm we simulate darts throws from a distribution as was done in [2].

The methodology and setup of these experiments will first be explained, followed by a presentation and discussion of the results.

A. Methodology

In our experiments we simulated agents with 45 skill levels, evenly spaced on the interval $[10, 170]$. Each agent had their skill estimated while they threw 60 darts at the target specified by the skill estimation method. The noise for all 60 dart throws was first sampled, so that each method saw the same variations from the chosen targets. Each method started with the same initial estimate of $\sigma = 50$, which is what was used in [2]. Each method also performed 10 steps of the EM algorithm, the same as was done in [2]. The following methods produced a new skill estimate after each set of 3 darts were thrown:

- **Center** – This method had the player aim at the center and used the skill estimation method from [2].
- **Triple 11** – This method had the player aim at the minimum entropy point, the triple 11, and used the MCEM skill estimation method presented in Section III.

- **Adaptive** – This method adapted the player's target after each set of 3 darts, based on the current estimate, following the adaptive method described in the previous section.

This experiment was repeated 100 different times, and the resulting percentage error from the MLE estimate using the true distances from the target to the dart locations was measured in each case. This error was averaged across all 45 skill levels for each experiment, and also across all 100 different experiments.

B. Experimental Results

Fig **??** shows the average percentage error across these experiments as a function of the number of dart throws that were observed. This shows the accuracy of each approach, as well as how quickly each approach converges to the MLE estimate. It is immediately apparent that the adaptive method performs the best, followed by the triple 11, and then the center. In order to understand the performance differences, we can look at the results in two ways.

1) Fixed number of Darts: First, we can compare the average error of each approach after a fixed number of darts. This can be done visually by drawing a vertical line across the plot. For example, at 45 darts the error is 5.5%, 6.7%, and 7.8% for the adaptive, triple 11, and center respectively. After 9 darts, the adaptive method has the lowest error. This means, that for any constant number of darts, the adaptive method yields the lowest error, on average.

2) Fixed error: The second way that we can look at the results is comparing the number of darts required to reach a fixed error level. This can be done visually be drawing a horizontal line at a level of allowable error. The points where this line intersects the results curves can then be used to see how many darts it would take each approach in order to obtain that level of accuracy. For example, to reach 10% error, 15, 18, and 24 darts are required for the adaptive, triple 11, and center respectively. The difference between the methods becomes more stark as the error tolerance decreases. If allowable error is decreased to 7.5% then the required darts increase to 24, 33, and 48 respectively. For comparison with the suggested procedure in [2], which required 50 darts to be thrown, our adaptive method can reach a similar error level with *only half as many darts*! This constitutes an enormous gain in the usability and applicability of this method, even in the limited setting where targets are prescribed for the players.

3) Consistency: Lastly we investigate the consistency of the different approaches. Following a similar structure to Figure **??**, Figure 7 shows the standard deviation of the percentage error across the 100 experiment repetitions, giving a good measure of how confident we can be with our estimate, and enabling us to provide a confidence interval when combined with the information in Figure **??**. The results show that the adaptive approach is far more consistent that the other single-target skill estimation approaches.

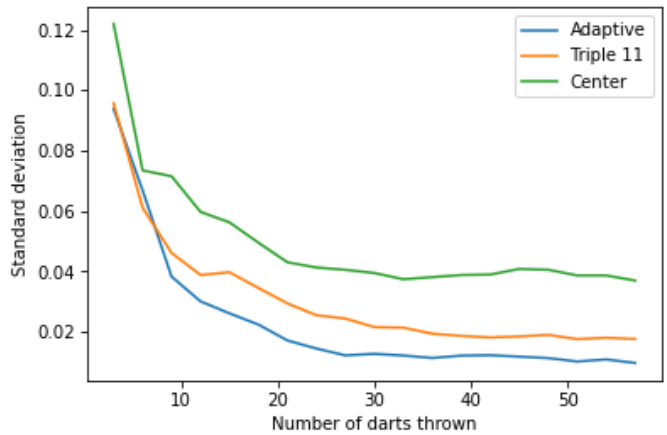

Fig. 7. The standard deviation of the percentage error for approaches

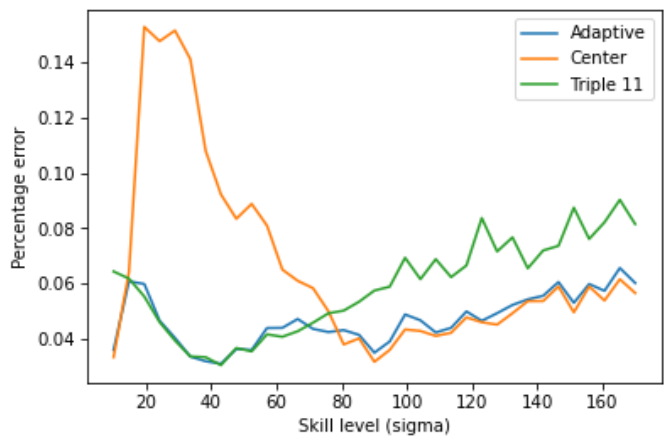

Fig. 8. Percent error after 60 darts for each approach

4) Performance for skill levels: Figure 8 shows the average percent error for each method after 60 darts as a function of the skill level of the agent. This figure shows that the center target approach suffers from high error for agents with a low σ values, while the triple 11 does worse than the center for agents with $\sigma > 74$. The benefit of the adaptive method is that it is able to maintain very close to the performance of the best of those two targets at every skill level.

No matter how the results are viewed, the proposed methods demonstrate clear advantages over the previous method from [2]. These improvements were made possible by MCEM and our reasoning about lower average entropy targets. The flexibility of the MCEM approach should lead to a wide variety of similarly successful and useful darts applications.

VII. Related Work

There are several strands of research related to the topics covered in this paper, including darts, execution skill, and modeling opponent capabilities and strategies in games.

Prior to [2], a few papers have explored the domains of darts. [6] focused on identifying an optimal strategy for darts

game, specifying where best to aim at different scores to minimize the number of darts required to end the game. This strategy was computed for a few different execution skill levels. Later work looked at a large set of actual dart throws from a few professional human dart players to explore the best type of model for execution noise in darts [7].

Execution skill, or precision, has been explored in a number of other contexts. In [8], the impact of varying execution skill on the performance of computer billiards agents was explored. Other work looked at estimating agent's execution skill in billiards and darts [9], [10]. In contrast to the current work, in those papers the precise dart location or action parameters were assumed to be observed, while the action targets were unobserved.

Determining the execution skill parameter of another agent is a type of opponent modeling, which has been explored extensively in many game contexts [11]–[17]. Popular opponent-modeling domains are games like poker, where there is unobserved information that, when combined with characteristics of the opponent, can have a big impact on how the game will unfold.

Our research builds on these foundation to further the understanding of modeling players in the game of darts, as well as understanding execution skill in darts and other physical game domains.

VIII. Conclusion

This paper presented a novel method to estimate execution skill: Monte Carlo EM for darts. This approach offers increased flexibility over the existing method, and this flexibility was shown to lead to a more efficient, accurate, and consistent adaptive skill estimation procedure.

The future extensions and applications of the MCEM approach are numerous. The Monte Carlo framework should allow for the easy use of other noise distributions, like asymmetric or skew Gaussian, to represent the execution error in darts. We also plan to use skill estimates to handicap players in darts, making games between different skilled players even and more exciting. Additionally, we aim to use the adaptive estimation procedure in human trials to explore both its usability and topics related to how humans make decisions and understand their own skill in games like darts.

Generalizations of this approach could provide skill estimates in domains beyond darts, with other target-based games being the easiest adaptations. Many of these domains will suffer from the same limitation that observed results fail to provide complete data about actions taken by an agent. This could include shooting, billiards, curling, golf, cornhole, and archery.

References

[1] A. E. Elo, *The rating of chessplayers, past and present*. Arco Pub., 1978.

[2] R. J. Tibshirani, A. Price, and J. Taylor, "A statistician plays darts," *Journal of the Royal Statistical Society: Series A (Statistics in Society)*, vol. 174, no. 1, pp. 213–226, 2011.

[3] "National dart association rules," https://ndadarts.com/nda-rules/, accessed: 2021-08-20.

[4] A. P. Dempster, N. M. Laird, and D. B. Rubin, "Maximum likelihood from incomplete data via the em algorithm," *Journal of the Royal Statistical Society: Series B (Methodological)*, vol. 39, no. 1, pp. 1–22, 1977.

[5] G. C. Wei and M. A. Tanner, "A monte carlo implementation of the em algorithm and the poor man's data augmentation algorithms," *Journal of the American statistical Association*, vol. 85, no. 411, pp. 699–704, 1990.

[6] D. Kohler, "Optimal strategies for the game of darts," *Journal of the Operational Research Society*, vol. 33, no. 10, pp. 871–884, 1982.

[7] H. S. Stern and W. Wilcox, "A statistician reads the sports pages: Shooting darts," *Chance*, vol. 10, no. 3, pp. 16–19, 1997.

[8] C. Archibald, A. Altman, and Y. Shoham, "Success, strategy and skill: an experimental study," in *Proceedings of the 9th International Conference on Autonomous Agents and Multiagent Systems*. International Foundation for Autonomous Agents and Multiagent Systems, 2010, pp. 1089–1096.

[9] C. Archibald and D. Nieves-Rivera, "Execution skill estimation," in *Proceedings of the 17th International Conference on Autonomous Agents and MultiAgent Systems*, ser. AAMAS '18. International Foundation for Autonomous Agents and Multiagent Systems, 2018, pp. 1859–1861.

[10] ——, "Bayesian execution skill estimation," *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 33, no. 01, pp. 6014–6021, Jul. 2019.

[11] N. Bard, D. Nicholas, C. Szepesvari, and M. Bowling, "Decision-theoretic clustering of strategies," in *Proceedings of the Fourteenth International Conference on Autonomous Agents and Multi-Agent Systems (AAMAS)*, 2015.

[12] N. Bard, M. Johanson, N. Burch, and M. Bowling, "Online implicit agent modelling," in *Proceedings of the Twelfth International Conference on Autonomous Agents and Multi-Agent Systems (AAMAS)*, 2013, pp. 255–262.

[13] N. Bard and M. Bowling, "Particle filtering for dynamic agent modelling in simplified poker," in *Proceedings of the Twenty-Second Conference on Artificial Intelligence (AAAI)*, 2007, pp. 515–521.

[14] N. Sturtevant, M. Zinkevich, and M. Bowling, "ProbMaxn: Opponent modeling in n-player games," in *Proceedings of the Twenty-First National Conference on Artificial Intelligence (AAAI)*, 2006, pp. 1057–1063.

[15] P. McCracken and M. Bowling, "Safe strategies for agent modelling in games," in *AAAI Fall Symposium on Artificial Multi-agent Learning*, October 2004.

[16] F. Schadd, S. Bakkes, and P. Spronck, "Opponent modeling in real-time strategy games." in *GAMEON*, 2007, pp. 61–70.

[17] D. Carmel and S. Markovitch, "Opponent modeling in multi-agent systems," in *International Joint Conference on Artificial Intelligence*. Springer, 1995, pp. 40–52.